

# CSE 5830: Probabilistic Graph Analysis

## Homework 3

Due 11:59PM, Wednesday 9/30/2026

Show all your work to get a full score.

### Problem 1: Noisy-AND (15 points)

A Noisy-AND model (independence of the causal mechanisms when all  $X$ s are 1) with binary random variables and leaky parameters  $(\lambda_0, \lambda_1, \dots, \lambda_K)$ .

#### (a) Expression for $P(Y = 0 \mid X_1, \dots, X_K)$ (10 points)

**Setup / convention.** Noisy-AND is the De Morgan dual of noisy-OR. In a deterministic AND gate, a single input  $X_i = 0$  is enough to force  $Y = 0$ , and  $Y = 1$  happens in only one way: every input equals 1.

| $X_1$ | $X_2$ | $Y = X_1 \wedge X_2$ |
|-------|-------|----------------------|
| 0     | 0     | 0                    |
| 0     | 1     | 0                    |
| 1     | 0     | 0                    |
| 1     | 1     | 1                    |

For each parent, introduce an independent mechanism variable  $Z_i$ , where  $Z_i = 1$  means “ $X_i$  forces  $Y = 0$ ”. An input that is present can never force an AND gate to 0, so

$$P(Z_i = 1 \mid X_i = 1) = 0, \quad P(Z_i = 1 \mid X_i = 0) = \lambda_i.$$

That is,  $\lambda_i$  is the probability that an absent input ( $X_i = 0$ ) succeeds in forcing  $Y = 0$ . The leak  $Z_0$  is always active, with  $P(Z_0 = 1) = \lambda_0$ : the probability that  $Y = 0$  even when every input is 1. Then  $Y = 1$  if and only if  $Z_0 = Z_1 = \dots = Z_K = 0$ .

**Derivation.** Since  $Y = 1$  requires that no mechanism fires, and the mechanisms are independent (independence of the causal mechanisms),

$$P(Y = 1 \mid X_1, \dots, X_K) = P(Z_0 = 0) \prod_{i=1}^K P(Z_i = 0 \mid X_i).$$

Each factor is

$$P(Z_i = 0 \mid X_i) = \begin{cases} 1, & X_i = 1, \\ 1 - \lambda_i, & X_i = 0, \end{cases}$$

so parents with  $X_i = 1$  contribute a factor of 1 and drop out. Hence

$$P(Y = 1 \mid X_1, \dots, X_K) = (1 - \lambda_0) \prod_{i: X_i=0} (1 - \lambda_i),$$

and taking the complement,

$$P(Y = 0 \mid X_1, \dots, X_K) = 1 - (1 - \lambda_0) \prod_{i: X_i=0} (1 - \lambda_i)$$

**Sanity check.** With no leak ( $\lambda_0 = 0$ ) and  $\lambda_i = 1$  for all  $i$ : if every  $X_i = 1$ , the product is empty (equal to 1), so  $P(Y = 0) = 0$ ; if any  $X_i = 0$ , the product contains a factor  $1 - 1 = 0$ , so  $P(Y = 0) = 1$ . This is exactly the deterministic AND gate.

### (b) Numerical case (5 points)

$K = 4$ ,  $\lambda_0 = 0$ ,  $\lambda_1 = 0.9$ ,  $\lambda_2 = 0.75$ ,  $\lambda_3 = 0.8$ ,  $\lambda_4 = 0.85$ .

Only  $X_2$  and  $X_3$  are off, so only  $\lambda_2$  and  $\lambda_3$  enter the product;  $X_1 = X_4 = 1$  contribute factors of 1. With no leak,  $1 - \lambda_0 = 1$ . Using the formula from (a):

$$\begin{aligned} P(Y = 0 \mid X_1 = 1, X_2 = 0, X_3 = 0, X_4 = 1) &= 1 - (1 - \lambda_0)(1 - \lambda_2)(1 - \lambda_3) \\ &= 1 - (1 - 0)(1 - 0.75)(1 - 0.8) \\ &= 1 - (1)(0.25)(0.2) \\ &= 1 - 0.05 \\ &= \boxed{0.95} \end{aligned}$$

**Sanity check.** Two inputs are off, and each has a high chance (0.75 and 0.8) of forcing  $Y = 0$ . So  $Y = 1$  requires both of those mechanisms to fail, which happens with probability only  $0.25 \times 0.2 = 0.05$ . A value of  $P(Y = 0)$  close to, but below, 1 is what we expect.

## Problem 2: Factor product (15 points)

Given factors:

| $X$ | $Y$ | $\phi_1(X, Y)$ | $Y$ | $Z$ | $\phi_2(Y, Z)$ |
|-----|-----|----------------|-----|-----|----------------|
| 1   | 1   | 0.8            | 1   | 1   | 0.2            |
| 1   | 2   | 0.5            | 1   | 2   | 0.2            |
| 2   | 1   | 0.5            | 2   | 1   | 0.9            |
| 2   | 2   | 0.6            | 2   | 2   | 1.0            |

(a)  $\psi(X, Y, Z) = \phi_1(X, Y) \phi_2(Y, Z)$  (10 points)

| $X$ | $Y$ | $Z$ | $\phi_1(X, Y) \cdot \phi_2(Y, Z)$ | $\psi(X, Y, Z)$ |
|-----|-----|-----|-----------------------------------|-----------------|
| 1   | 1   | 1   | _____ . _____                     | _____           |
| 1   | 1   | 2   | _____ . _____                     | _____           |
| 1   | 2   | 1   | _____ . _____                     | _____           |
| 1   | 2   | 2   | _____ . _____                     | _____           |
| 2   | 1   | 1   | _____ . _____                     | _____           |
| 2   | 1   | 2   | _____ . _____                     | _____           |
| 2   | 2   | 1   | _____ . _____                     | _____           |
| 2   | 2   | 2   | _____ . _____                     | _____           |

(b)  $\psi(X, Z \mid Y = 1)$  (5 points)

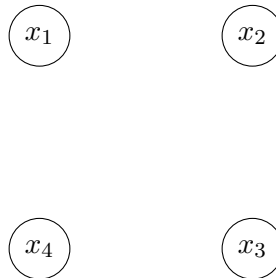
[TODO: Explain in one or two sentences what operation you are applying (factor reduction to the context  $Y = 1$ ) and whether you normalize.]

| $X$ | $Z$ | $\psi(X, Z \mid Y = 1)$ |
|-----|-----|-------------------------|
| 1   | 1   | _____                   |
| 1   | 2   | _____                   |
| 2   | 1   | _____                   |
| 2   | 2   | _____                   |

**Problem 3: Pairwise Markov network (10 points)**

$$p(\mathbf{x}) \propto \phi(x_1, x_2) \phi(x_2, x_3) \phi(x_3, x_4) \phi(x_4, x_1)$$

(a) Draw the network (2 points)



(b) Conditionals in terms of  $\phi$  (8 points)

[TODO: Show the derivation once in full for  $p(x_1 \mid x_2, x_4)$  (write the conditional as a ratio of joints, then cancel), then use the same pattern for the rest.]

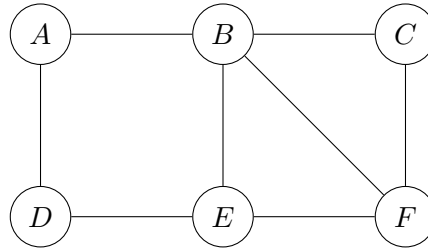
$$p(x_1 \mid x_2, x_4) = [\text{TODO:}]$$

$$p(x_2 \mid x_1, x_3) = [\text{TODO:}]$$

$$p(x_3 \mid x_2, x_4) = [\text{TODO:}]$$

$$p(x_4 \mid x_1, x_3) = [\text{TODO:}]$$

**Problem 4: Markov network structure (20 points)**



(a) All factors that factorize over the graph (6 points)

Edges: **[TODO: list them]**

Cliques: **[TODO: list every clique (single nodes, edges, triangles, ...)]**

Maximal cliques: **[TODO: ]**

$$P(A, B, C, D, E, F) = \frac{1}{Z} [\text{TODO: product of factors}]$$

(b) Gibbs distribution as a pairwise MN (6 points)

$$P(A, B, C, D, E, F) = \frac{1}{Z} [\text{TODO: product of pairwise factors}], \quad Z = [\text{TODO:}]$$

(c) Local Markov independencies (6 points)

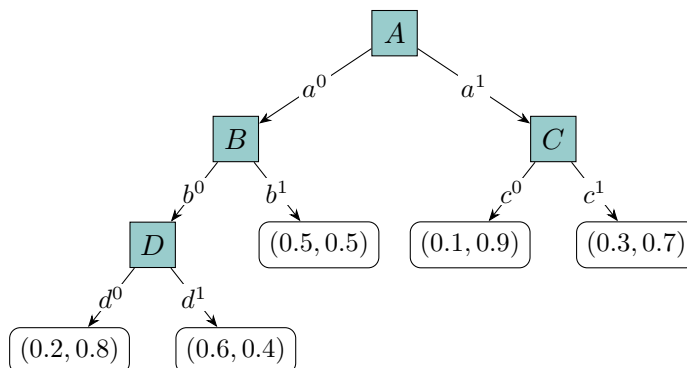
| Node | Neighbors (Markov blanket) | Local independence |
|------|----------------------------|--------------------|
| A    | _____                      | <b>[TODO: ]</b>    |
| B    | _____                      | <b>[TODO: ]</b>    |
| C    | _____                      | <b>[TODO: ]</b>    |
| D    | _____                      | <b>[TODO: ]</b>    |
| E    | _____                      | <b>[TODO: ]</b>    |
| F    | _____                      | <b>[TODO: ]</b>    |

(d) Is  $A \perp F \mid \{D, C\}$ ? (2 points)

**[TODO: Answer yes/no, and justify with separation: either exhibit an active path from A to F that avoids {D, C}, or argue that every path passes through {D, C}.]**

## Problem 5: Tree-structured CPD (10 points)

Leaves list  $(P(e^0), P(e^1))$ .



(a)  $P(E = e^0 \mid a^0, b^1, c^1, d^1)$

Path through the tree: **[TODO: ]**      Answer: \_\_\_\_\_

(b)  $P(E = e^1 \mid a^1, b^1, c^1, d^1)$

Path through the tree: **[TODO: ]**      Answer: \_\_\_\_\_

(c)  $P(E = e^1 \mid a^0, b^0, c^0, d^0)$

Path through the tree: **[TODO: ]**      Answer: \_\_\_\_\_

(d) Is  $E \perp_c B \mid a^0$ ?

**[TODO: Yes/no + justification using the tree.]**

(e) Is  $E \perp_c C \mid d^1$ ?

**[TODO: Yes/no + justification using the tree.]**

### Use of Ai

In this assignment, I used Ai to help me understand the assignment give me hints for each question as well as helping whenever I got stuck in a question. Also used AI to write and debug my tex code. However, I solved all the question of the assignment on my own.